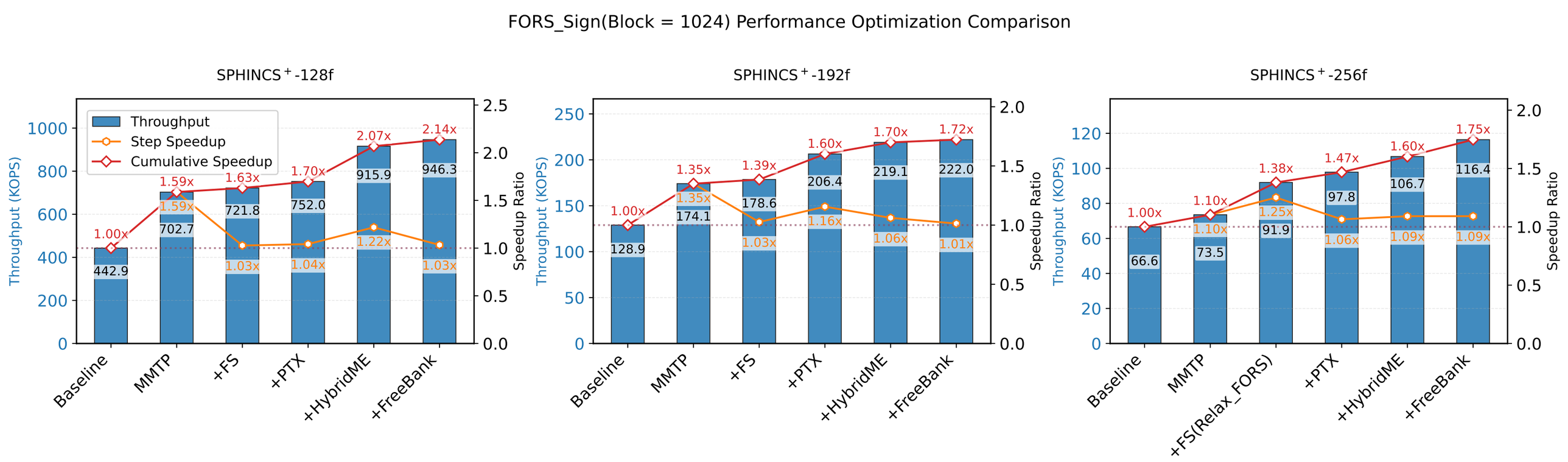
FORS_Sign(Block = 1024) Performance Optimization Comparison
SPHINCS⁺-128f
SPHINCS⁺-192f
SPHINCS⁺-256f
Throughput
Step Speedup
Cumulative Speedup
Throughput (KOPS)
Speedup Ratio
Baseline
MMTP
+FS
+FS(Relax_FORS)
+PTX
+HybridME
+FreeBank
442.9
702.7
721.8
752.0
915.9
946.3
128.9
174.1
178.6
206.4
219.1
222.0
66.6
73.5
91.9
97.8
106.7
116.4
1.00x
1.59x
1.63x
1.70x
2.07x
2.14x
1.35x
1.39x
1.60x
1.70x
1.72x
1.10x
1.38x
1.47x
1.60x
1.75x
1.03x
1.04x
1.22x
1.16x
1.06x
1.01x
1.25x
1.09x